AI Chatbot / Question Answering Application

Using Ollama as the Local AI Model and Streamlit as the User Interface

1. Project Title

Local AI Chatbot using Ollama and Streamlit

2. Project Objective

The objective of this project is to build a simple AI chatbot / question-answering application that runs an AI language model locally using Ollama. A Streamlit web interface is used so that the user can enter a question or prompt and view the generated response in the browser. The application demonstrates how a local large language model can be connected to a simple Python interface.

3. What is Ollama?

Ollama is a tool for running large language models locally on a computer. It provides a simple way to download, manage, and interact with supported models through a local service/API. In this project, Ollama acts as the AI model layer: the user's prompt is sent to the selected local model, and the generated answer is returned to the Streamlit application.

4. What is Streamlit?

Streamlit is a Python framework for creating interactive web applications quickly. It is used in this project to create the chatbot interface, accept user input, send the prompt to Ollama, and display the AI-generated response without requiring a separate frontend framework.

5. Ollama Model Used

Model: Llama 3.2 (local Ollama model)
Note: If a different Ollama model was installed for the actual demonstration, replace "Llama 3.2" with that model name in this section before submission.

6. Technologies Used

Technology	Purpose
Python	Main programming language
Ollama	Runs the local AI language model
Llama 3.2	Generates answers to user prompts
Streamlit	Creates the interactive web interface

7. Working of the Application

- The user starts the Ollama service and makes the selected model available locally.
- The user runs the Streamlit Python application.
- The Streamlit page provides a text input area for entering a question or prompt.
- When the user submits the prompt, the Python application sends it to the Ollama local model.
- The Ollama model processes the prompt and generates a response.
- The response is returned to the Streamlit application and displayed on the web page.

8. Basic System Flow

User → Streamlit Interface → Python Application → Ollama Local Model → Generated Response → Streamlit Interface → User

9. Test Prompts

The application can be tested with at least five different prompts. The following prompts cover general knowledge, programming, education, explanation, and creative use cases.

No.	Test Prompt
1	What is Artificial Intelligence?
2	Explain Python in simple words.
3	What is the difference between AI and Machine Learning?
4	Give five tips for preparing for a technical interview.
5	Explain how a chatbot works in simple language.

10. Application Features

- Local AI processing using Ollama.
- Simple and user-friendly Streamlit interface.
- Accepts natural-language questions and prompts.
- Displays generated answers directly in the browser.
- Does not require a cloud AI API key for the local Ollama model.
- Can be extended later with chat history, prompt templates, document question answering, or other models supported by Ollama.

11. Advantages

- Easy to build and demonstrate using Python.
- AI model can run locally on the user's computer.
- Simple interface suitable for beginners.
- The model can be changed to another supported Ollama model.
- Useful as a foundation for study assistants, notes generators, and other AI applications.

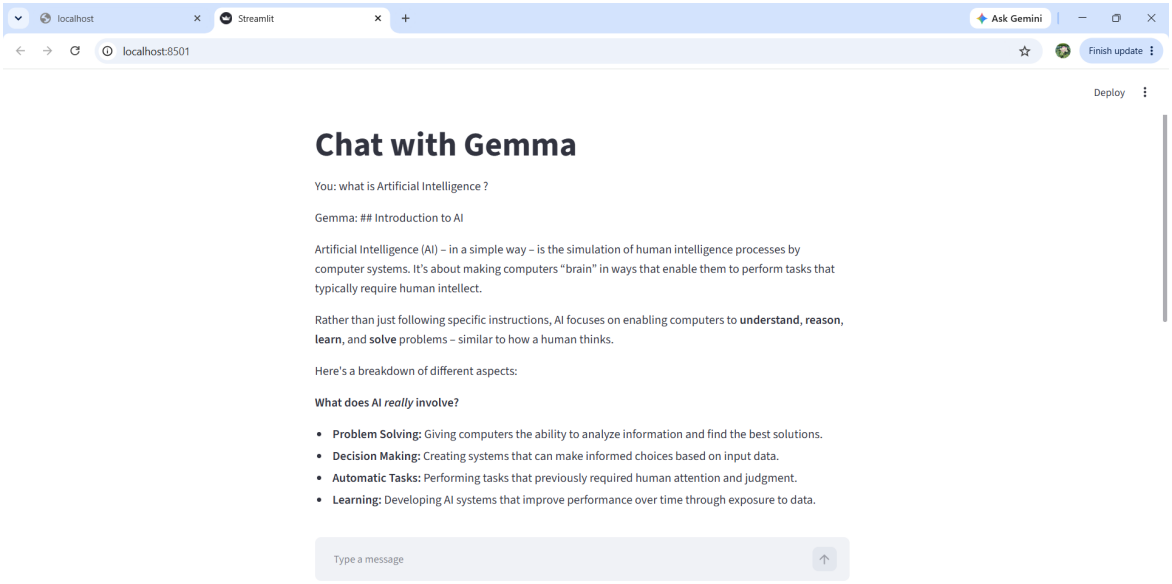
12. Limitations

The response quality and speed depend on the selected local model and the computer's available CPU/GPU/RAM resources. A local model may also have knowledge limitations and may occasionally produce incorrect information. The application therefore should not be treated as an authoritative source for high-stakes decisions.

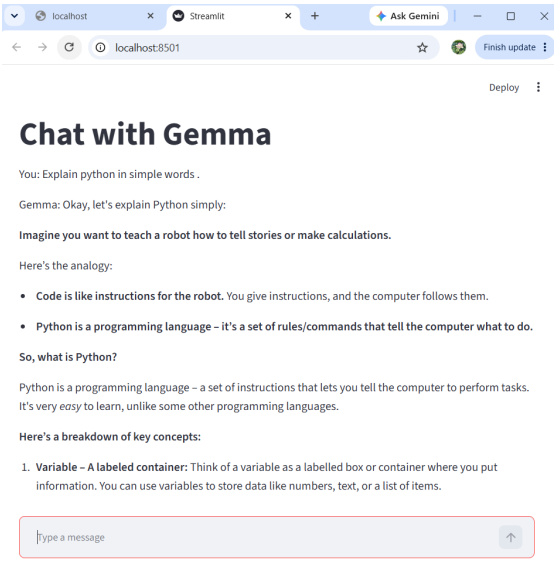
13. Conclusion

This project demonstrates a basic local AI question-answering application using Ollama and Streamlit. Ollama provides the local language-model capability, while Streamlit provides a simple browser-based interface for entering prompts and viewing responses. The project provides a practical introduction to connecting a Python application with a locally running AI model and can be extended into more advanced AI assistants in the future.

Appendix: Working Application Screenshots

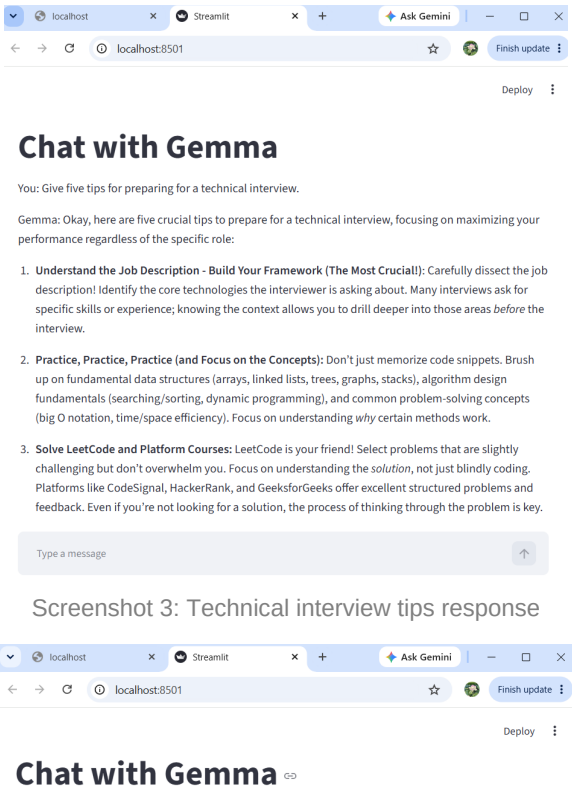


Screenshot 1: Chatbot answering "What is Artificial Intelligence?"

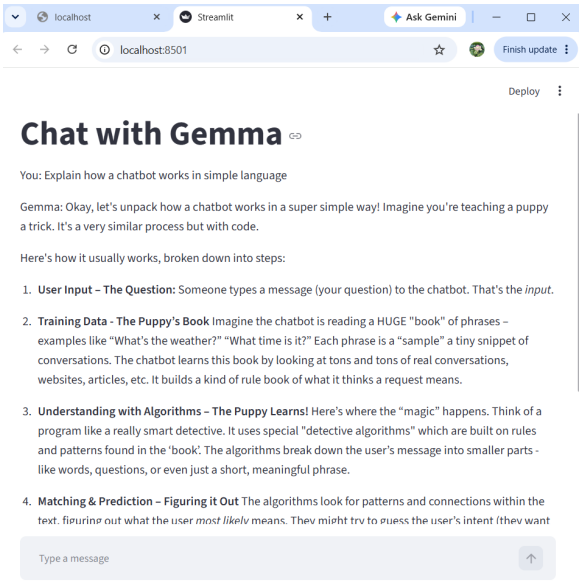


Screenshot 2: Chatbot explaining Python in simple words

Working Application Screenshots (continued)

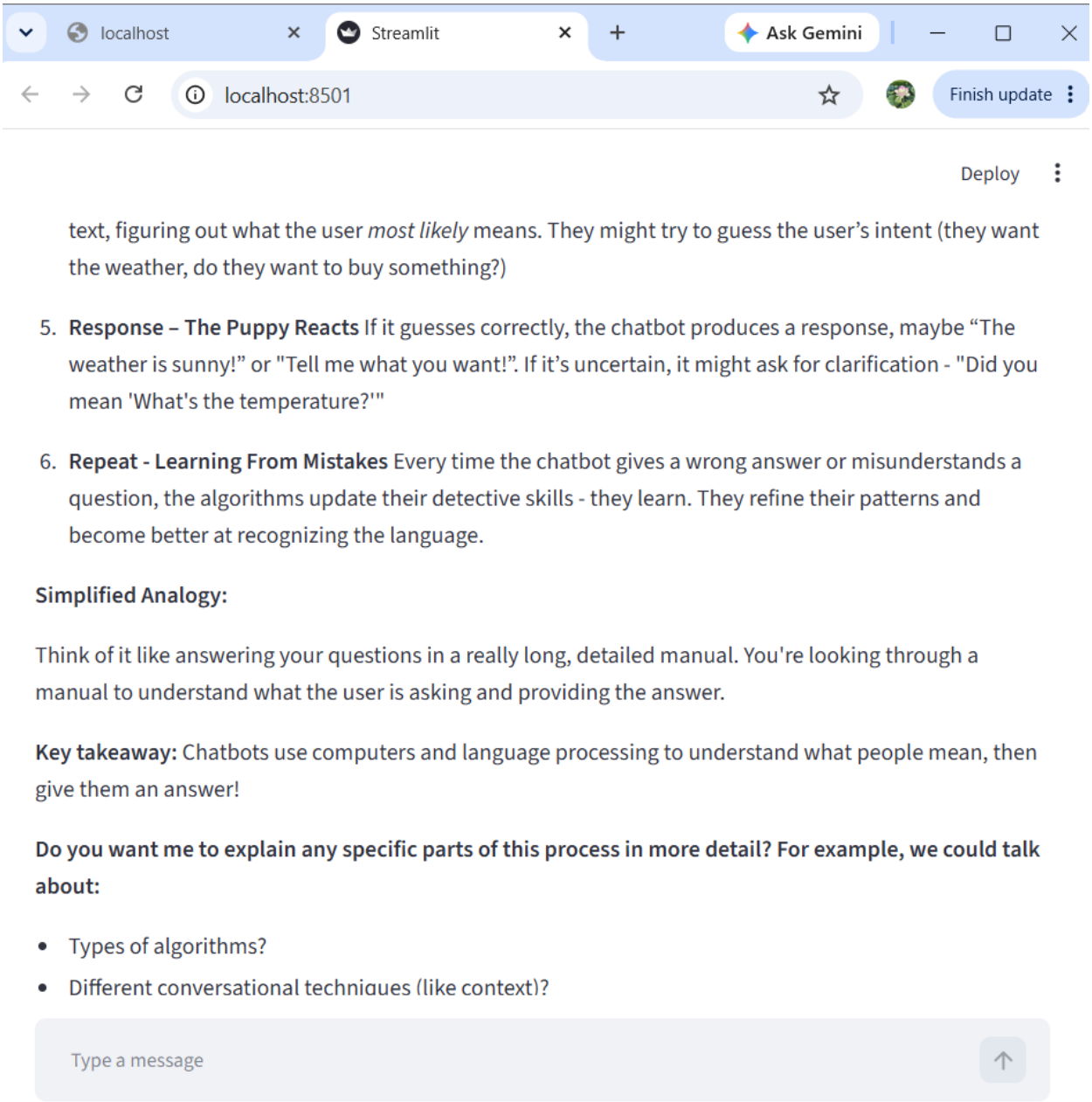


Screenshot 3: Technical interview tips response



Screenshot 4: Explanation of how a chatbot works

Working Application Screenshots (continued)



Screenshot 5: Continued chatbot response and conclusion